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Interaction of *gem*-Difluorinated Iodides with Silyl Enol Ethers Mediated by Photoredox Catalysis

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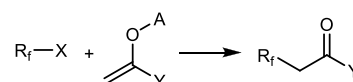
Abstract. A method for the synthesis of β,β -difluorinated ketones by coupling of α,α -difluorinated iodides with silyl enol ethers is described. The reaction is performed in the presence of iridium(III) photocatalysts under the irradiation with 400 nm light emitting diodes.

Keywords: fluorine; photocatalysis; silyl enol ethers; radical reactions; fluoroalkylation

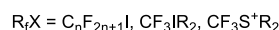
The introduction of fluorine-containing groups into organic molecules can modify their biological properties.^[1] Compounds bearing *gem*-difluorinated methylene fragment have gained increasing attention in medicinal chemistry owing to isosteric relationship of the CF₂-fragment to ethereal oxygen.^[2] Correspondingly, in recent times, there has been a surge of work aimed at the development of efficient methods for their preparation.^[3-5] While the CF₂-fragment can be readily created from the keto group by deoxofluorination reaction,^[6] the required conditions are typically quite harsh narrowing the scope of tolerant functional groups. Furthermore, application of this method for the synthesis of *gem*-difluorinated compounds having an additional carbonyl group would be problematic.

Alkylation of ketones at the α -position by a fluorinated reagent constitutes an attractive general pathway towards selectively fluorinated products (Scheme 1). Correspondingly, reactions of enol derivatives with perfluoroalkyl iodides^[7-9] or perfluoroalkyl iodine(III)^[10] or sulfur reagents^[11] have been extensively investigated.^[12-14] However, similar alkylation processes using reagents containing only two fluorine atoms (that is RCF₂I, where R is alkyl group) have not been reported. This is, primarily, because these reagents are, generally, difficult to access. Moreover, fluoroalkylation reactions of enolates frequently proceed by ion-radical mechanism involving reductive carbon-iodine bond

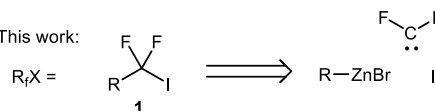
breaking, and, as a consequence, partially fluorinated reagents are less reactive compared to perfluorinated counterparts.



Previous work:



This work:



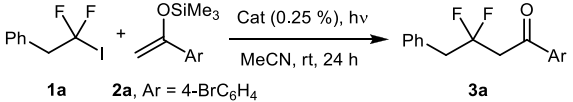
Scheme 1. Fluoroalkylation of enol derivatives.

Recently we reported a straightforward method to synthesize *gem*-difluorinated iodides **1** by difluorocarbene insertion into carbon-zinc bond followed by iodination^[15] (Scheme 1). Herein we report an application of these iodides for the alkylation of silyl enol ethers under mild photoredox conditions.

Iodide **1a** was selected as a model substrate and its reaction with silyl enol ether **2a** (1.3 equiv) derived from 4-bromoacetophenone was evaluated (Table 1). In the presence of *fac*-Ir(ppy)₃ (0.25 mol %) upon irradiation with 400 nm light emitting diodes, product **3a** was formed in only 14% yield. Notable increase in yield was achieved by addition of 1.2 equiv of propylene oxide (PO), which serves as a scavenger of trimethylsilyl iodide produced as a side product. The addition of 0.2 equiv of triphenylphosphine provided further improvement leading to the product in 87% isolated yield. Another iridium photocatalyst, [Ir(ppy)₂(dtbpy)]PF₆, was equally effective, whereas a ruthenium complex gave reduced yield. The phosphine can play a two-fold role. Thus, the phosphine can effect reductive quenching of the

photocatalyst excited state affording a strongly reducing species.^[16] On the other hand, the phosphine can prevent the photocatalyst from deactivation by traces of oxygen. We propose that the latter mode is more likely, since phosphine oxide is observed by GC-MS analysis of reaction mixtures.

Table 1. Optimization studies.



Cat. ^{a)}	Additive (equiv) ^{b)}	Yield, % ^{c)}
Ir(ppy) ₃	–	14
Ir(ppy) ₃	PO (1.2)	79
Ir(ppy) ₃	PO (1.2), PPh ₃ (0.2)	87 ^{d)}
Ir(ppy) ₃	PPh ₃ (0.2)	30
Ir(ppy) ₃	AcONa (1.2), PPh ₃ (0.2)	32
[Ir(ppy) ₂ (dtbpy)]PF ₆	PO (1.2), PPh ₃ (0.2)	87
Ru(bipy) ₃ (BF ₄) ₂	PO (1.2), PPh ₃ (0.2)	69

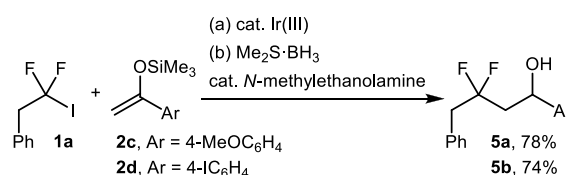
^{a)} For Ir-catalyst, 400-nm LED; for Ru-catalyst, blue LED.

^{b)} PO – propylene oxide. ^{c)} Determined by ¹⁹F NMR of reaction mixtures with 4-fluorotoluene as an internal standard. ^{d)} Isolated yield.

Under the optimized conditions a series of iodides **1** were coupled with silyl enol ethers **2** (Table 2). Iodides **1** bearing aryl, ester or phosphonate groups gave good yields of β,β -difluoroketones **3** after acidic workup and column chromatography. Silyl enol ethers derived from various aromatic and heteroaromatic ketones worked well. However, 1-trimethylsilyloxy cyclohexene (silyl enol ether derived from cyclohexanone) was markedly less reactive affording less than 15% yield of the expected product in reaction with iodide **1a** (¹⁹F NMR control of the reaction mixture). This was surprising given that enol ether **2g** derived from α -tetralone gave reasonable yield of ketone **3g** (entry 6).

Though these fluoroalkylation processes are typically quite clean, the reaction mixtures typically contain small amounts of by-products resulting from elimination of hydrogen fluoride from ketones **3**.^[17,18] For acyclic ketones, the amount of elimination by-products did not exceed 5 %, whereas in reaction of α -tetralone derived enol ether **2g**, about 10 % of the corresponding β -fluoroenone was observed by ¹⁹F NMR. Furthermore, for compounds **3c,d**, purification on silica gel gave products in diminished yields compared to the yields determined by NMR of reaction mixtures (entries 2 and 3), which may also be associated with dehydrofluorination. To avoid the loss of products, crude samples of ketones **3c,d** were reduced with borane-dimethylsulfide complex catalyzed by *N*-methylethanolamine furnishing alcohols **5a,b** in good overall yields (Scheme 2).

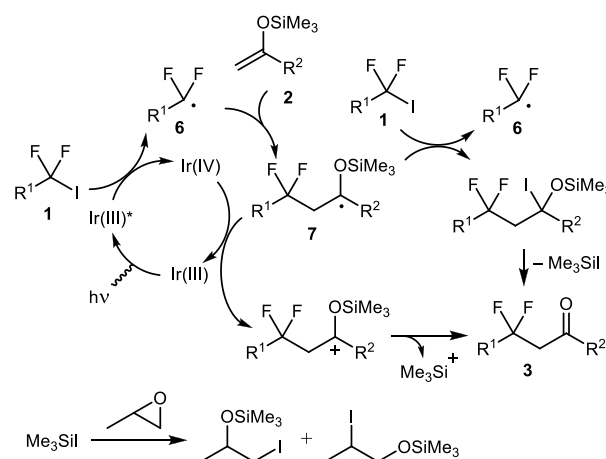
Proposed mechanism for the photoredox reaction is shown in Scheme 3. Catalytic cycle starts from the single electron reduction of iodide **1** by excited state of



Scheme 2. Synthesis of alcohols **5**.

the photocatalyst. To confirm the feasibility of this step, we measured the reduction peak potential of **1a** to be -1.11 V (vs SCE), which is noticeably lower than the potential of -1.73 V for Ir(III)*/Ir(IV) couple.^[19,20] Redox potentials of other reaction components were also evaluated (see Supporting Information for CV curves). In particular, propylene oxide was electrochemically inactive within wide range in both cathodic and anodic regions. For silyl enol ether **2a**, the oxidation potential of +1.75 V was found suggesting that it cannot be oxidized by photoexcited Ir(ppy)₃ (Ir(III)*/Ir(IV) +0.31 V). The oxidation of triphenylphosphine (*E*_{ox} +0.77 V)^[16] by iridium complex is also unfavorable.

The fluorinated radical **6** adds at the electron rich double bond of the enol ether generating silyloxy-substituted radical **7**. The latter can be oxidized by iridium(IV) leading to cationic intermediate, which would rapidly expel trimethylsilyl cation leading eventually to iodosilane. Alternatively, radical **7** can itself get engaged in a chain process by abstracting the iodine atom from the starting substrate **1** with the generation of fluorinated radical **6**. The role of propylene oxide is to trap highly Lewis acidic iodosilane. Products of the interaction of iodosilane with propylene oxide, isomeric silylated iodohydrins, were identified by GC-MS analysis.^[21]



Scheme 3. Proposed mechanism.

In summary, a convenient method for the synthesis of β,β -difluoroketones by fluoroalkylation of silyl enol ethers is described. The reaction is performed under photoredox conditions using *gem*-difluorinated iodides as sources of fluorinated free radicals.

Table 2. Fluoroalkylation of silyl enol ethers.

Entry	Iodide	Enol Ether	Product	Yield of 3 , % ^{a)}
1		1a	2b	3b 87
2			2c	3c 70 (91) ^{b)}
3 ^{c)}			2d	3d 69 (85) ^{b)}
4 ^{d)}			2e	3e 67
5			2f	3f 87
6			2g	3g 69
7			2h	3h 61
8		1b	2c	3i 87
9			2i	3j 79
10		1c	2j	3k 90
11 ^{e)}			2k	3l 63
12			2l	3m 54 (67) ^{b)}
13 ^{d)}		1d	2b	3n 70
14		1e	2b	3o 75

^{a)} Isolated yield. ^{b)} Determined by ¹⁹F NMR. ^{c)} Reaction time 72 h. ^{d)} [Ir(ppy)₂(dtbpy)]PF₆ was used as catalyst. ^{e)} Reaction time 48 h.

Experimental Section

Synthesis of ketones 3 (General Procedure 1).

A test tube equipped with stirring bar was charged with triphenylphosphine (26 mg, 0.1 mmol, 0.2 equiv), evacuated and backfilled with argon. MeCN (1.1 mL) was added and the solution was degassed by evaporation of about 10% of the solvent under vacuum followed by filling the tube with argon. Then, iodide **1** (0.5 mmol, 1 equiv), silyl enol ether **2** (0.66 mmol, 1.33 equiv), propylene oxide (42 μ L, 0.6 mmol, 1.2 equiv) and photocatalyst [for **3e,m**, $[\text{Ir}(\text{ppy})_3]$ (0.8 mg, 1.25 μ mol); for **3c,d**, $[\text{Ir}(\text{ppy})_2(\text{dtbbpy})]\text{PF}_6$ (1.1 mg, 1.25 μ mol)] were successively added. The reaction vessel was irradiated with 400 nm LEDs [for **3a-c,e-j,l-n**, 24 h; for **3d**, 72 h; for **3k**, 48 h]; during irradiation the mixture was cooled with room temperature water. For the workup, the mixture was treated with 1M aq. HCl (2 mL) and extracted with hexanes/methyl *tert*-butyl ether (1/1, 3 $\times$ 4 mL). The combined extracts were filtered through Na_2SO_4 , concentrated under vacuum, and the residue was purified by column chromatography on silica gel.

Synthesis of alcohols 5a,b (General Procedure 2).

The crude material of compounds **3c,d** (obtained according to General Procedure 1 after workup) was dissolved in THF (1 mL). Then, $\text{Me}_2\text{S}\cdot\text{BH}_3$ (190 μ L, 2 mmol, 4 equiv) and *N*-methylethanolamine (8 μ L, 0.1 mmol, 0.2 equiv) were added and reaction mixture was stirred for 2 hours at r.t. The mixture was quenched with water (1 mL) and extracted with EtOAc (3 $\times$ 3 mL). The combined extracts were filtered through Na_2SO_4 , concentrated under vacuum, and the residue was purified by column chromatography on silica gel.

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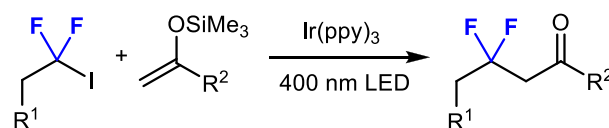
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